SE444 Makeup Mid-Term I – Answer Key & Grading Guide (Fall 2025)

Part I: Multiple Choice (4 pts)

Option	Q1 (1pt)	Q2 (1pt)	Q3 (1pt)	Q4 (1pt)
A	**X**	**X**	**X**	
B		**X**		**X**
C	**X**		**X**	
D	**X**		**X**	**X**
E		**X**		

Answers: Q1: A,C,D — Q2: A,B,E — Q3: A,C,D — Q4: B,D **Grading:** 1pt if all correct; 0.5pt if 1 error; 0pt if 2+ errors.

Q1: Heuristics
- A: Euclidean admissible with $\sqrt{2}$ diagonal ✓
- B: Manhattan inadmissible (overestimates)
- C: max dominates both ✓
- D: Weighted A* trades optimality for speed ✓
- E: With cost-1 diagonals, interpretation varies

Q2: Alpha-Beta
- A: $\alpha \geq \beta$ triggers pruning ✓
- B: MIN returns $\beta = 5$ ✓
- C: Pruned values can't change root
- D: This is alpha cutoff, not beta
- E: Different ordering affects when cutoff found ✓

Q3: Exploration vs Exploitation
- A: High temp too long prevents convergence ✓
- B: 100% elitism doesn't fully eliminate exploration
- C: High T_0 = max exploration ✓
- D: Restarts explore, climbing exploits ✓
- E: $h(n)$ role in A* mischaracterized

Q4: Local Optima
- A: Hill-climbing CAN mitigate plateaus
- B: Crossover combines optima ✓
- C: Stochastic HC not guaranteed
- D: Slow SA cooling has guarantees ✓
- E: Restarts don't guarantee global opt

Part II: A* Search (6 pts)

Q II.1: Formulation (1 pt)

State: Permutations of [0,1,2,3]
Initial: $S_0 = [3, 1, 0, 2]$
Actions: Swap blank with adjacent
Goal: State = [1,2,3,0]
Cost: 1 per move
Rubric: 0.5pt state/initial; 0.25pt actions; 0.25pt goal

Q II.2: A* Trace (3 pts)

Solution Path (3 moves):
$[3, 1, 0, 2] \xrightarrow{UP} [0, 1, 3, 2] \xrightarrow{R} [1, 0, 3, 2] \xrightarrow{D} [1, 2, 3, 0]$
Key f-values:
- $S_0 = [3, 1, 0, 2]$: $g = 0, h = 3, f = 3$
- $S_1 = [0, 1, 3, 2]$: $g = 1, h = 2, f = 3$
- $S_3 = [1, 0, 3, 2]$: $g = 2, h = 2, f = 4$
- Goal $[1, 2, 3, 0]$: $g = 3, h = 0, f = 3$

Rubric: 1pt correct f-values; 1pt frontier; 1pt solution path

Q II.3: Analysis (2 pts)

(a) Heuristics (1pt):
- Manhattan admissible? Yes (each tile needs $\geq$ Manhattan distance moves)
- Which expands fewer? Manhattan (dominates misplaced: $h' \geq h$)

(b) Greedy vs A* (1pt):
- Greedy optimal? No (ignores path cost)
- Greedy fewer: when h leads directly to goal
- Greedy more: when h misleads to dead-ends

Rubric: 0.5pt per sub-question

Part III: Genetic Algorithm (5 pts)

Q III.1: Representation (1 pt)

(a) Chromosome: [T1H1, T2H1, T1H2, T3H1]

(b) Search space: $6^4 = 1296$
(c) Advantage: Intuitive, valid by construction
Disadvantage: No natural distance metric

Q III.2: Fitness (2 pts)

S_1 = **[T1H1, T1H2, T2H1, T2H2]:**
- Hard: Smith clash at T1 → +100
- Soft: E1-E2 at T1 (25) + E3-E4 at T2 (30) → +40
- **Total: 140**

S_2 = **[T1H1, T2H1, T1H1, T3H2]:**
- Hard: E1,E3 at T1H1 → +100
- Soft: no shared students at same time → 0
- **Total: 100**

Fitter: S_2 (100 ¡ 140)

Q III.3: GA Operators (2 pts)

(a) Selection (0.5pt):
Tournament size 5 ¿ size 2 pressure
Higher pressure → more exploitation, premature convergence risk

(b) Crossover (1pt):
P1=[T1H1,T1H2,T2H1,T2H2], P2=[T2H1,T3H1,T1H2,T2H1]
2-point after pos 1,3:
- C1=[T1H1,T3H1,T1H2,T2H2] → penalty=100
- Result: C1 better than both parents (140, 200)

(c) Mutation (0.5pt):
- Too high (50%): random search, destroys building blocks
- Too low (0.1%): premature convergence
- Recommended: 1-5% (or $\approx 1/L$ per gene)